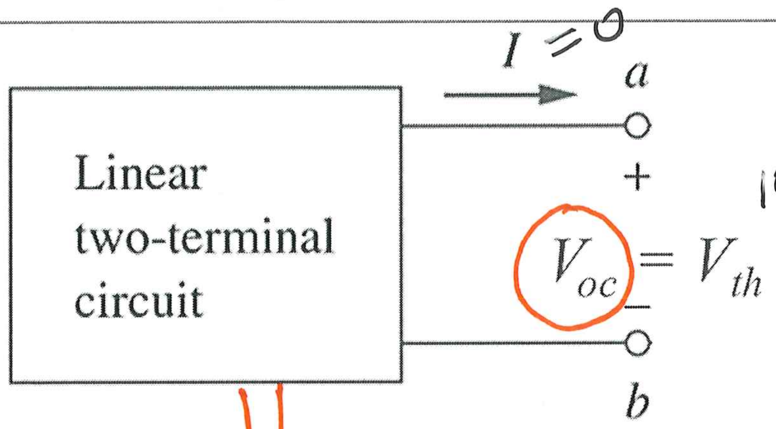
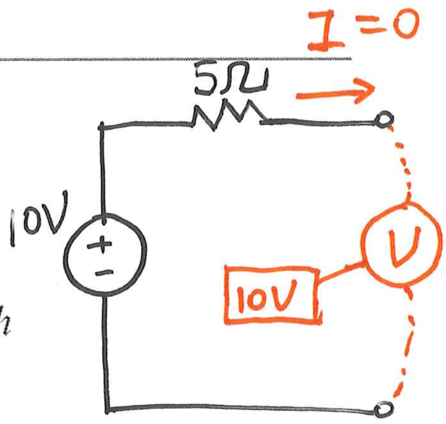
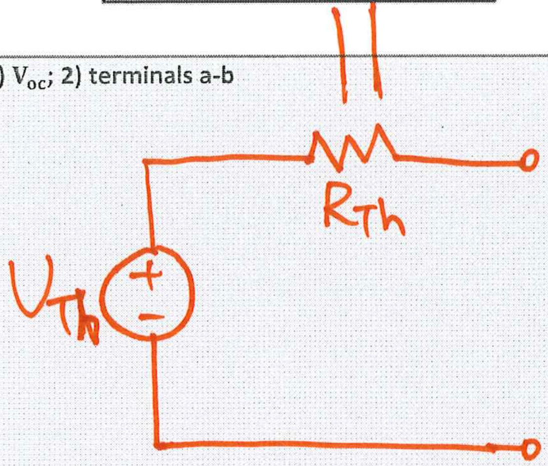
L#15: Thévenin's Theorem



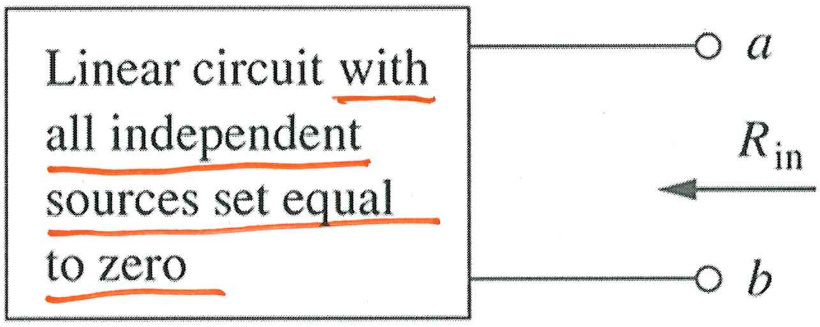


1) V_{oc}; 2) terminals a-b



$V_{Th} = 10V$

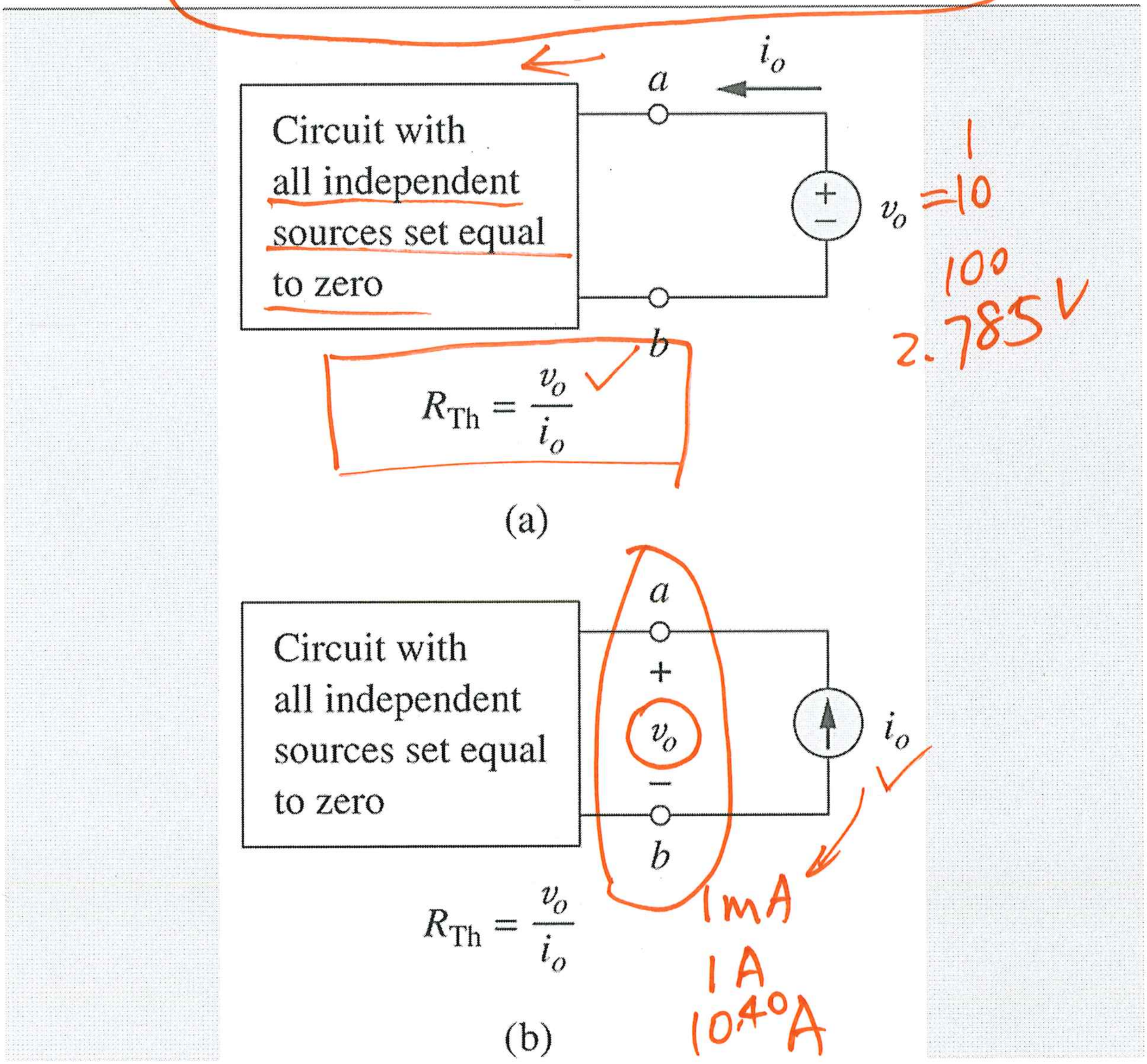
$R_{Th} = 5\Omega$



$$R_{\text{Th}} = R_{\text{in}}$$

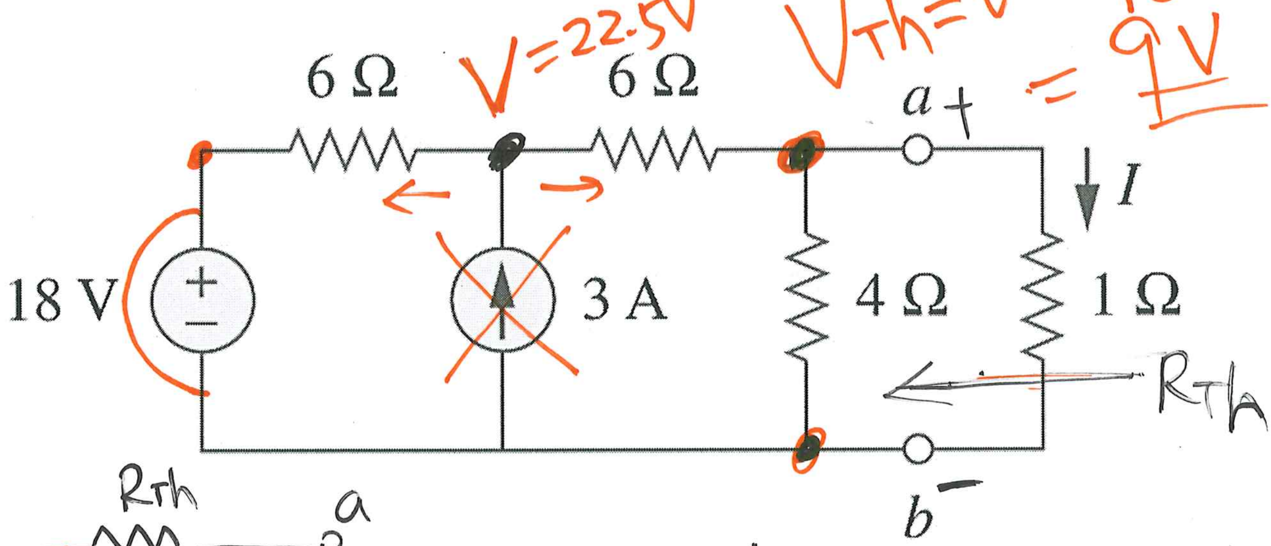
Method #1: Zeroing of Sources:
If the circuit contains no dependent sources.

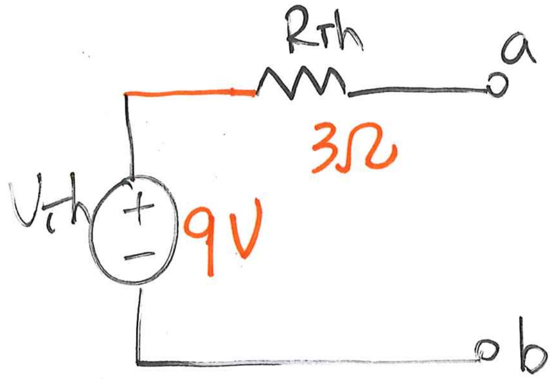
Method #2: Application of test sources, if the circuit does contain dependent sources.



Example #1: a) Determine Thévenin equivalent circuit; 2) Determine I.

$V_{Th} = V \cdot \frac{4}{10} = 9V$





$V_{Th} = V_{ab}$

$$3 = \frac{V-18}{6} + \frac{V}{10}$$

$$90 = 5V - 90 + 3V$$

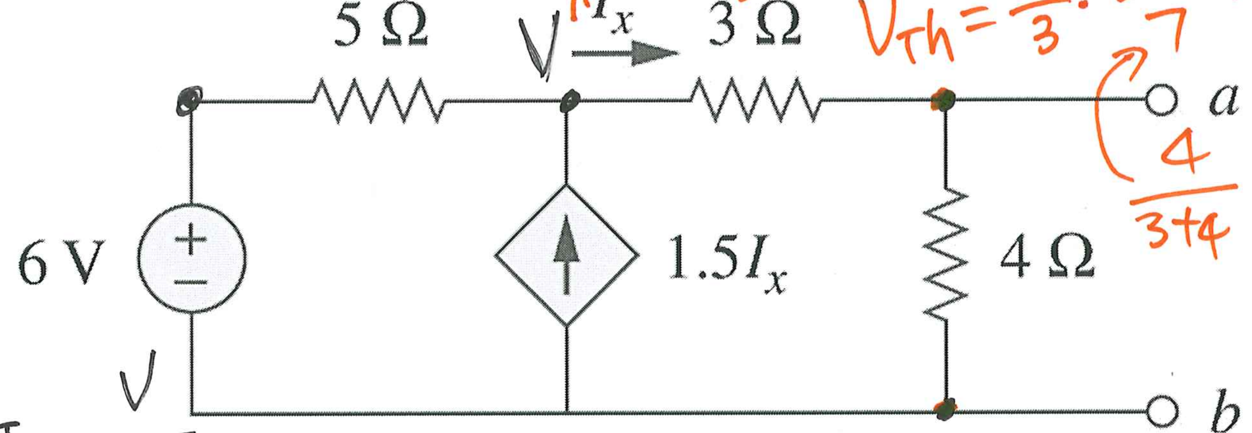
$$\Rightarrow 8V = 180 \Rightarrow V = 22.5V$$

$$R_{Th} = 4 \parallel (6+6) = \frac{48}{16} = 3\Omega.$$

Example #2: Determine the Thévenin equivalent circuit to the left of terminals a-b.

$\frac{28}{3}V$

$V_{Th} = \frac{28}{3} \cdot \frac{4}{7} = \frac{16}{3}V.$



$\frac{4}{3+4}$

$I_x = \frac{V}{7}$

Use 3 approaches for R_{th}

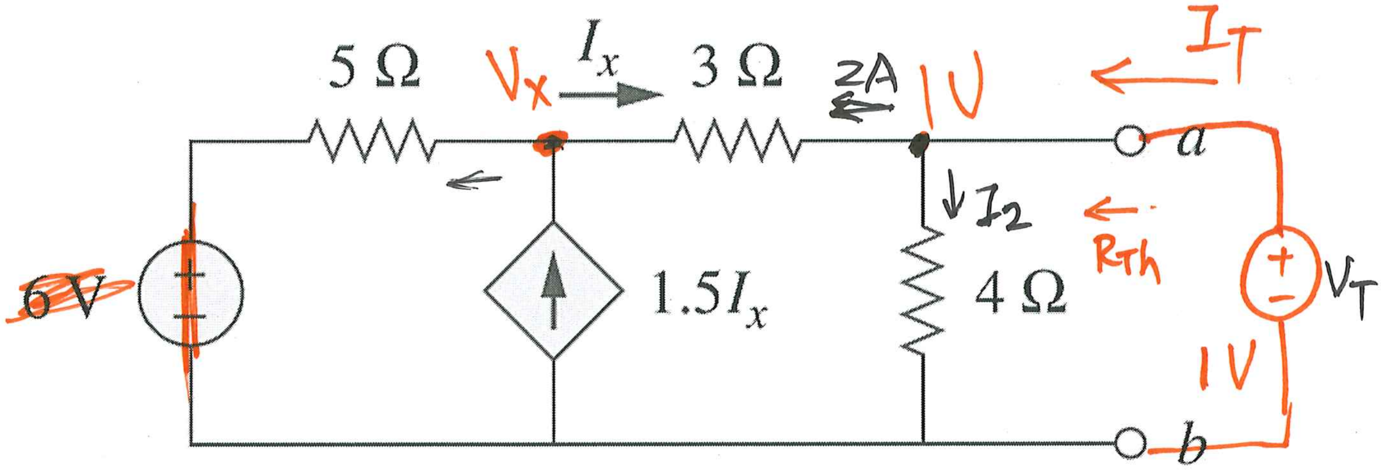
$V_{Th} = V_{ab}$, $\frac{V-6}{5} + \frac{V}{7} = 1.5 I_x = 1.5\frac{V}{7}$

$\frac{V-6}{5} = \frac{V}{14} \Rightarrow 14V - 84 = 5V$

$\Rightarrow 9V = 84$,

$V = \frac{84}{9} = \boxed{\frac{28}{3}} = 9.\dot{3}V$

$V_{Th} = \frac{16}{3} V.$



Add voltage source

$$1.5 I_x = \frac{V_x}{5} + \frac{V_x - 1}{3}, \quad I_x = \frac{V_x - 1}{3}$$

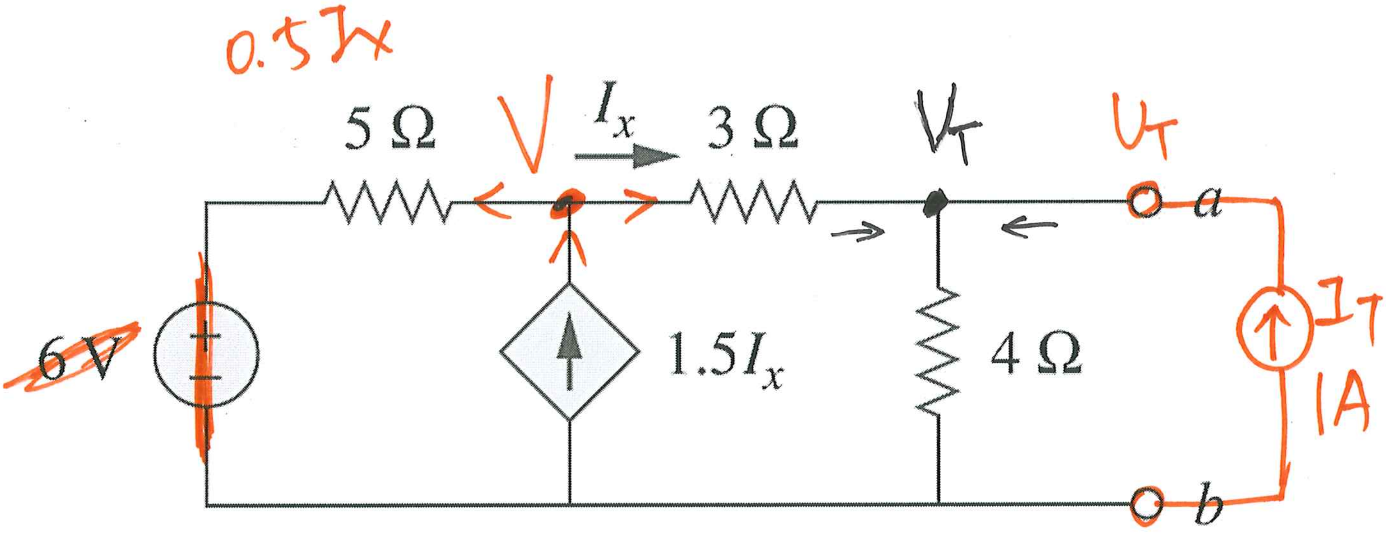
$$\cancel{0.5}\,\frac{V_x - 1}{6} = \frac{V_x}{5} \qquad 6V_x = 5V_x - 5$$

$$V_x = -5V.$$

$$I_x = \frac{-5-1}{3} = -2A, \quad I_2 = \frac{1}{4} = 0.25A$$

$$I_T = 2A + 0.25A = 2.25A = \frac{9}{4} A.$$

$$R_{Th} = \frac{V_T}{I_T} = \frac{1V}{9/4\,A} = \frac{4}{9}\,\Omega.$$



Add current source

$$0.5 I_x = \frac{V}{5} = 0.5 \cdot \frac{V - V_T}{3}$$

$$\frac{V}{5} = \frac{V - V_T}{6} \qquad (1) \Rightarrow 6V = 5V - 5V_T$$

$$\Rightarrow V = -5V_T$$

$$\frac{V - V_T}{3} + 1 = \frac{V_T}{4} \qquad (2)$$

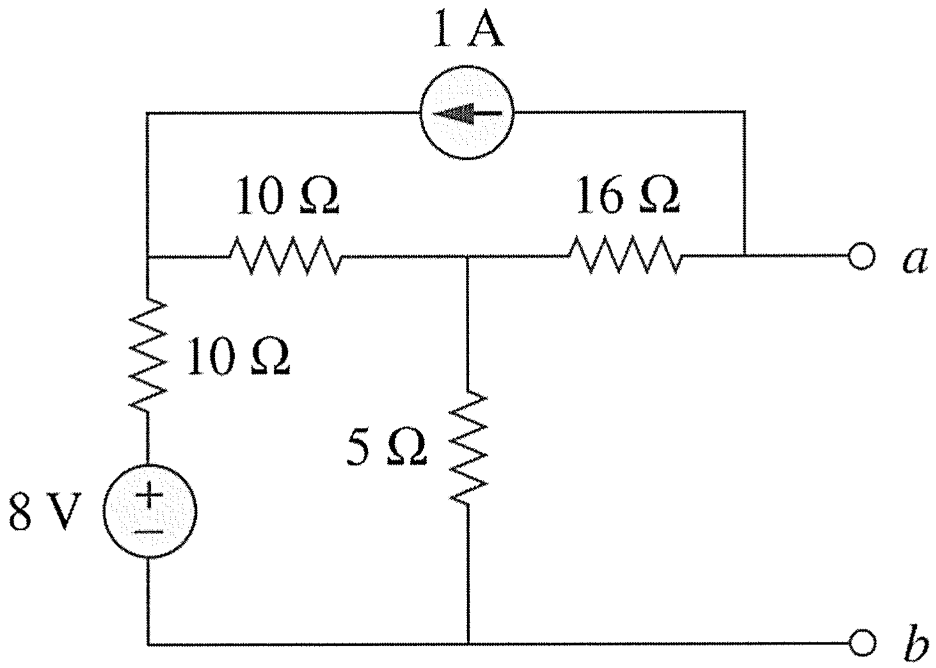
$$\frac{-5V_T - V_T}{3} + 1 = \frac{V_T}{4}$$

$$1 = \frac{V_T}{4} + 2V_T = \frac{9}{4} V_T$$

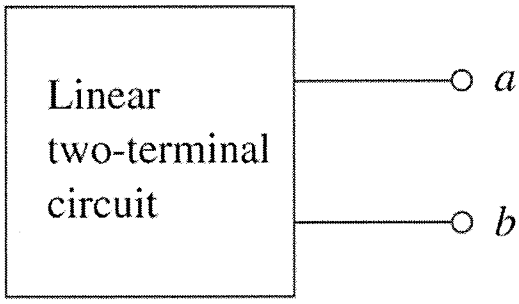
$$\Rightarrow V_T = \frac{4}{9} V \Rightarrow$$

$$R_{Th} = \frac{V_T}{I_T} = \frac{4/9\, V}{1\, A} = \frac{4}{9}\, \Omega.$$

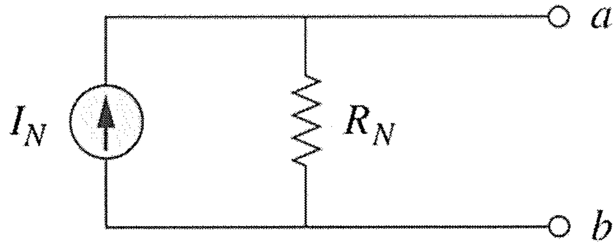
Example #3: Determine the Thévenin equivalent circuit.



L#16: Norton's Theorem



(a)

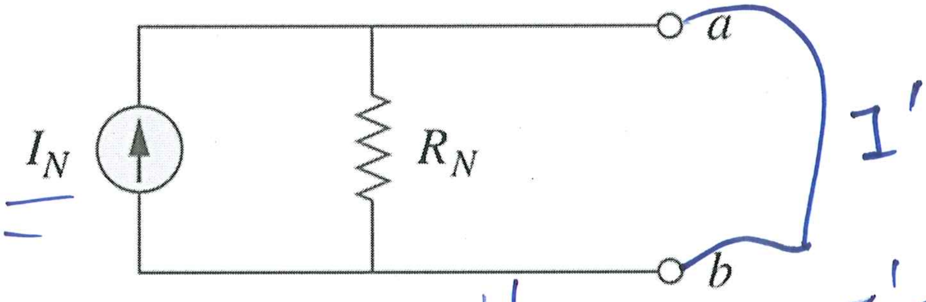


(b)

I_N: Norton equivalent current

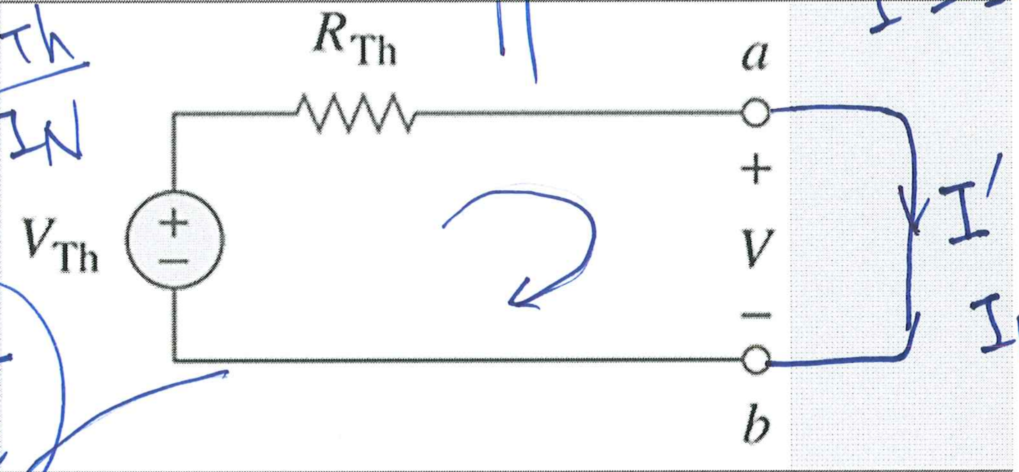
R_N: Norton equivalent Resistance

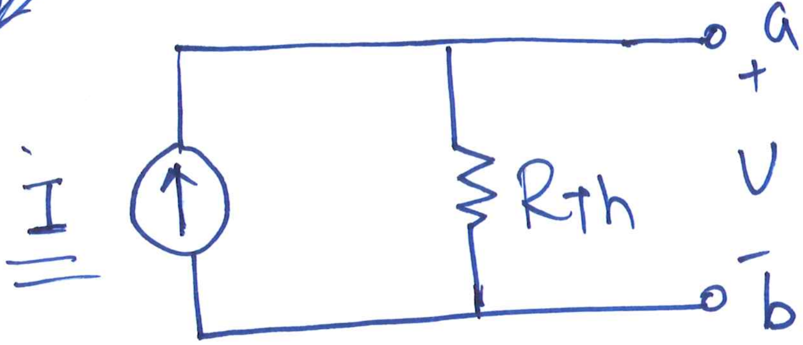
Norton's Theorem and Thévenin's Theorem



$I' = I_N$

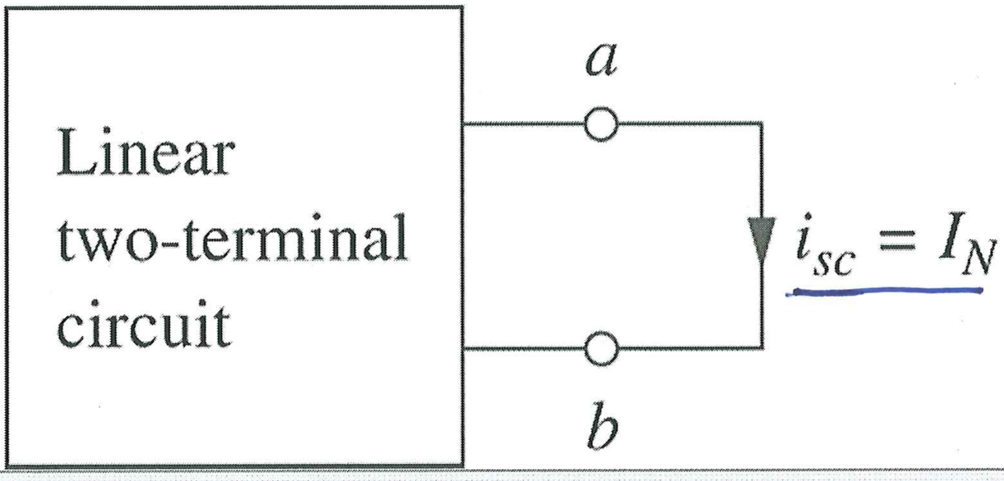
$R_{Th} = \frac{V_{Th}}{I_N} = \frac{V_{oc}}{I_{sc}}$





$I = \frac{V_{Th}}{R_{Th}}$, $\cancel{A}$ $I = I_N$, $R_{Th} = R_N$

Determine I_N



Norton equivalent current is the short circuit current

Determine R_N

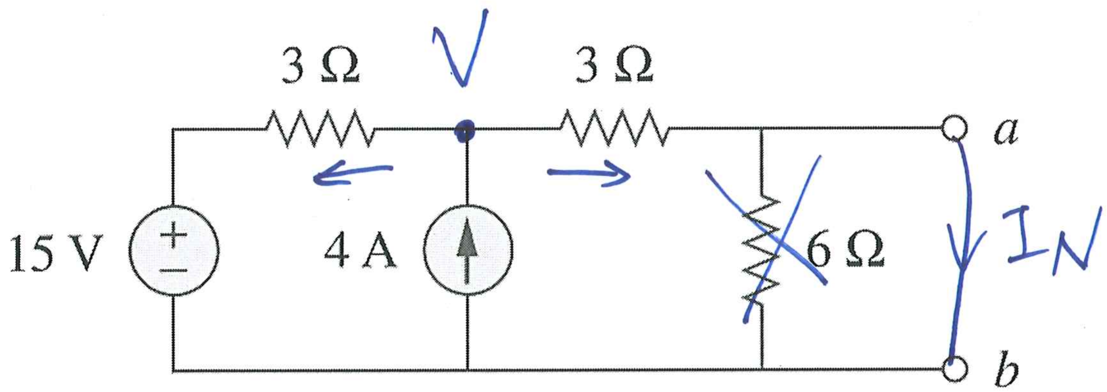
Method #1: Zero ind. sources.

Method #2: by testing A, V

Method #3:

- ***$R_N = R_{th} = V_{oc} / i_{sc}$*** $= V_{Th} / I_N$
- ***Works for circuits with or without dependent sources.***

Example #1: Determine the Norton equivalent circuit to the left of terminals a-b.



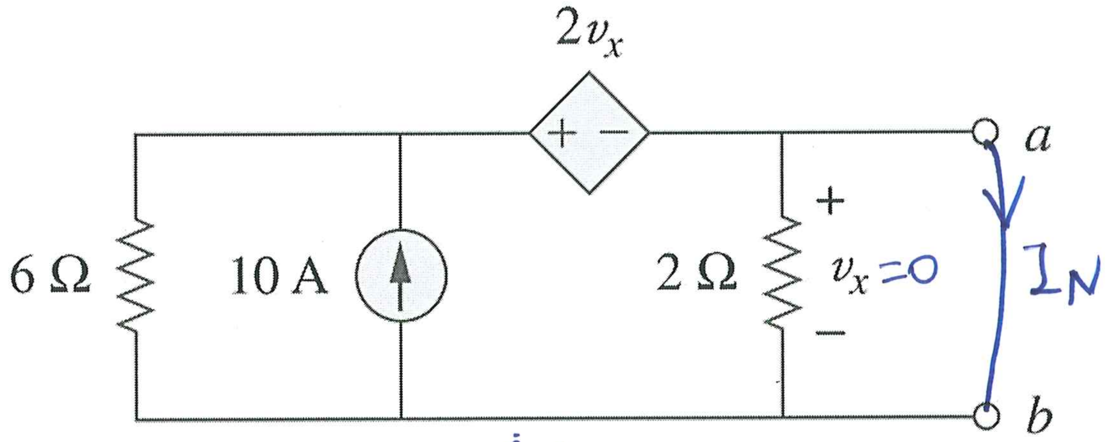
$$R_N = 6 \parallel (3+3) = 3\,\Omega.$$

$$4A = \frac{V-15}{3} + \frac{V}{3}$$

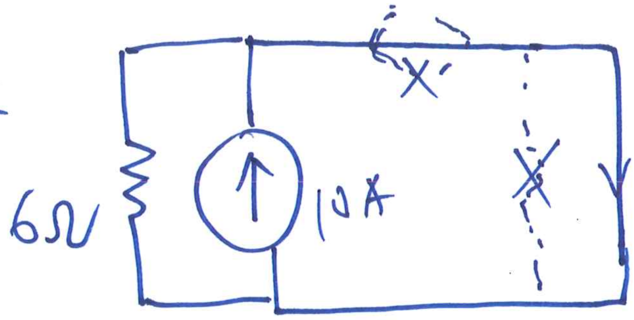
$$12 = 2V - 15 \Rightarrow V = 13.5$$

$$I_N = \frac{V}{3} = 4.5A$$

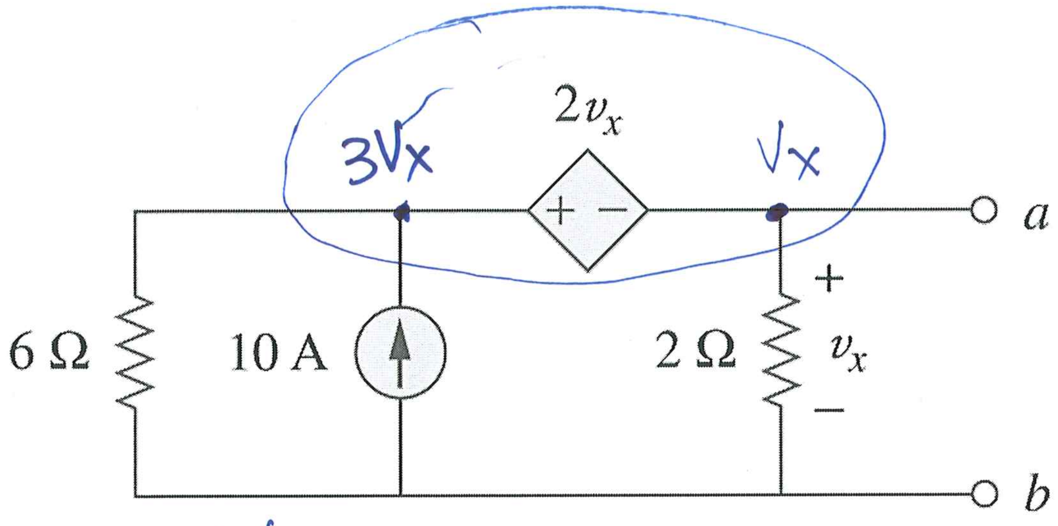
Example #2: Determine the Norton equivalent circuit to the left of terminals a-b.



I_N =



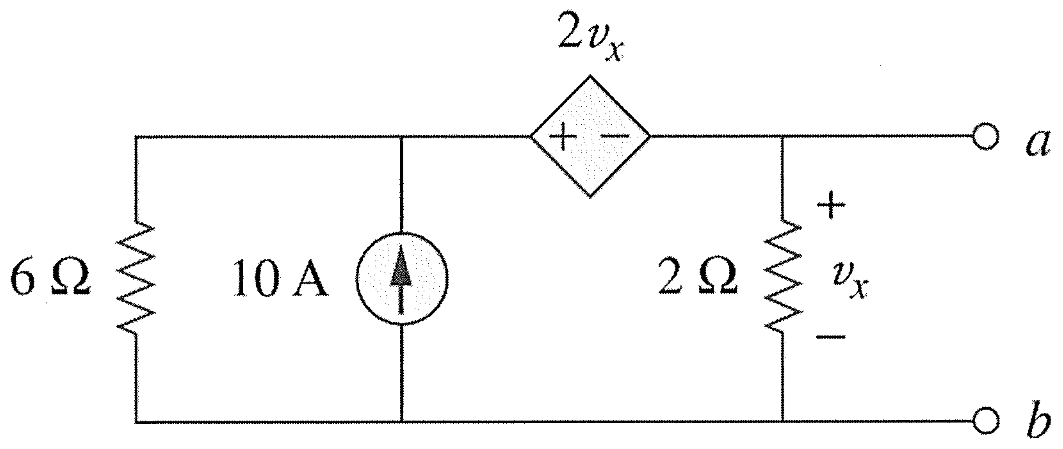
$\Rightarrow I_N = 10A$



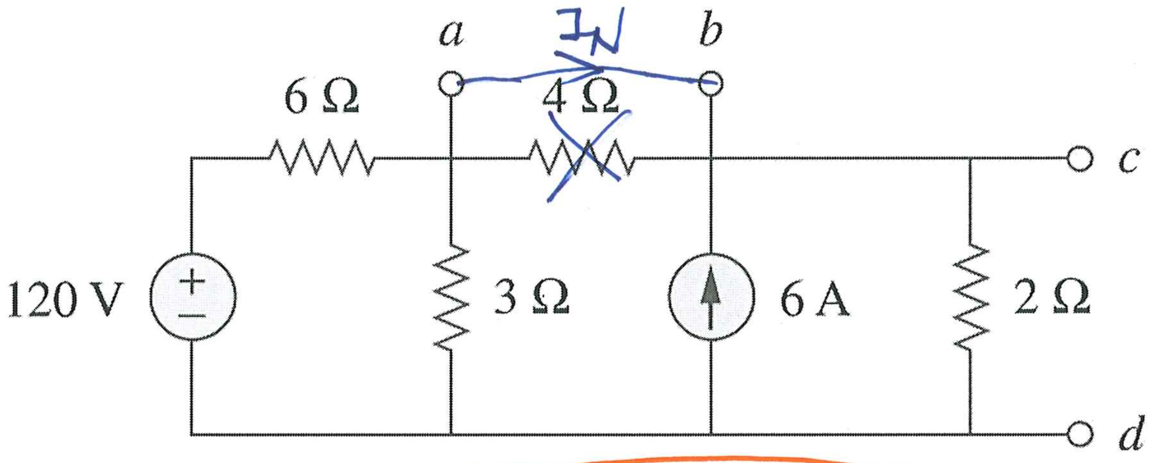
$$R_N = \frac{V_{oc}}{I_N}, \quad I_N = 10A.$$

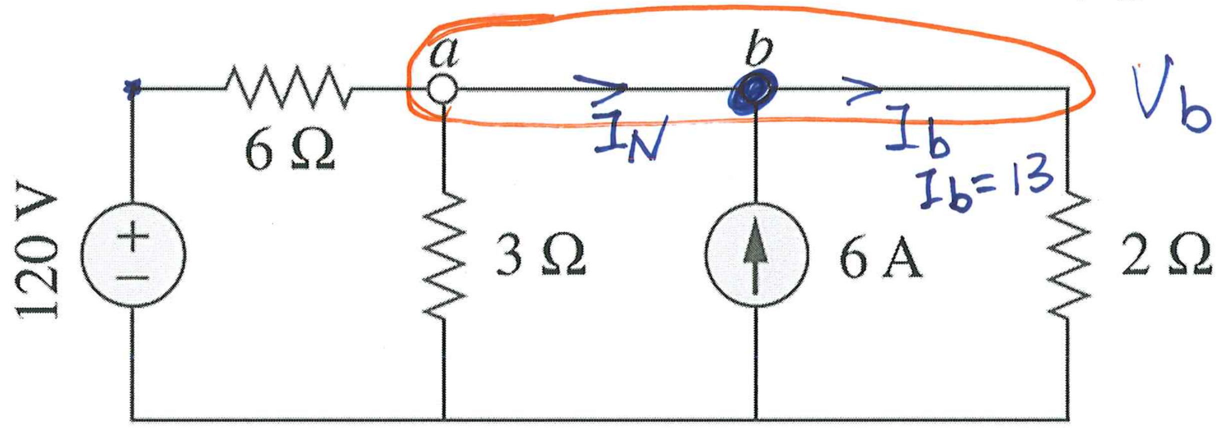
$$10 = \frac{3V_x}{6} + \frac{V_x}{2} \Rightarrow V_x = 10V.$$

$$R_N = \frac{V_{oc}}{I_N} = \frac{V_x}{I_N} = \frac{10V}{10A} = 1\Omega.$$


$2v_x$
+ −
a
$6\ \Omega$
$10\ \text{A}$
$2\ \Omega$
+
v_x
−
b

Example #3: Determine the Norton equivalent circuit. from a-b

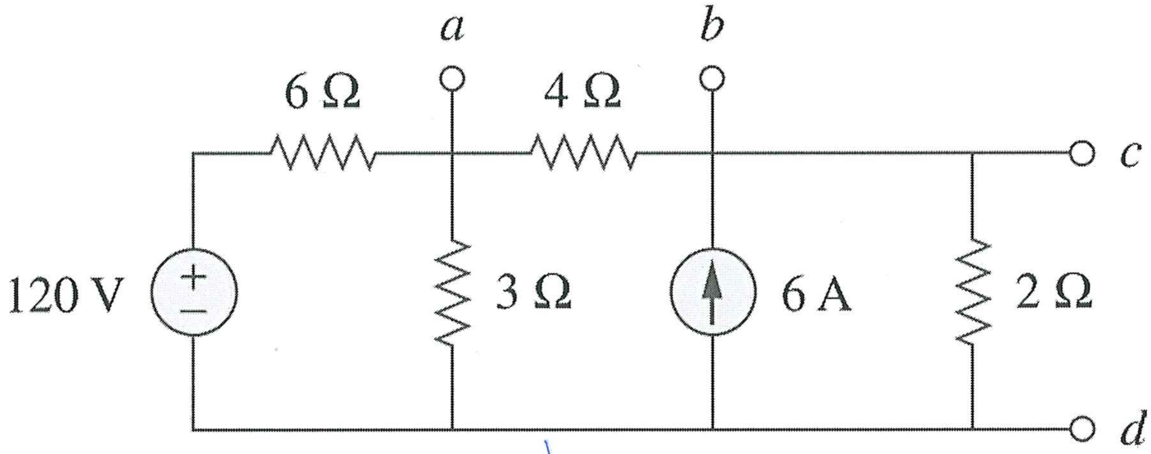


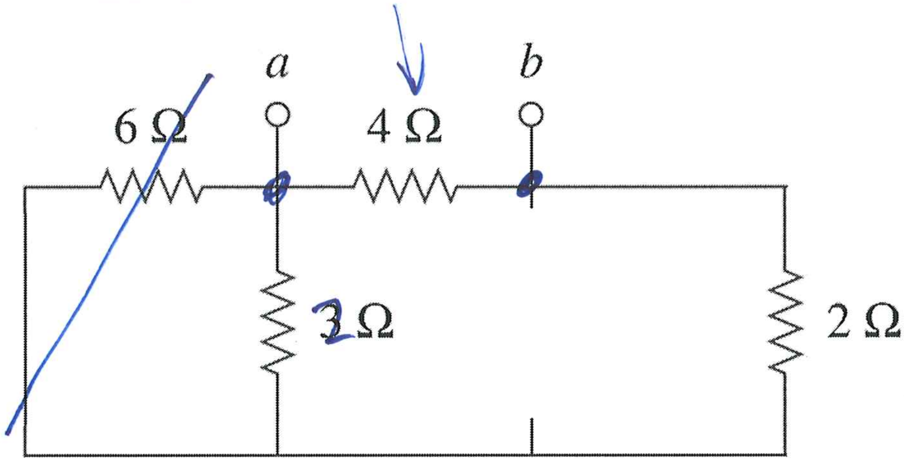


$$\frac{V_b - 120}{6} + \frac{V_b}{3} + \frac{V_b}{2} = 6$$

$$\frac{V_b}{6} + \frac{V_b}{3} + \frac{V_b}{2} = 26 \Rightarrow V_b = 26.$$

$$I_N = I_b - 6 = 7A$$





$$\frac{6 \times 3}{6+3} = 2$$

$$R_N = 4 \,||\, (2+2) = 2\Omega.$$